

Finite-Time Geometric Heat Flow for Fast and Feasible Trajectory Optimization Through Contact

Ram Vasudevan*

Department of Robotics

Department of Mechanical Engineering

University of Michigan

Ann Arbor, MI, USA

Abstract

Trajectory optimization for high-dimensional robotic systems remains too slow to serve as a real-time primitive, and this gap widens further when contact renders the dynamics hybrid. This talk presents the maturation of the Affine Geometric Heat Flow (AGHF) program described at last year’s meeting into a complete theory, solver, and benchmark suite for fast, dynamically feasible trajectory generation. The AGHF deforms an arbitrary curve in state space into a dynamically feasible trajectory by following the gradient flow of a constrained action functional; a realization theorem guarantees that the flow’s output can be tracked by admissible inputs. We solve this flow with a fixed-time stable proximal-point augmented Lagrangian method whose inner problems converge in finite time, paired with an exact-Hessian Newton method that exploits the banded structure of the collocated flow to achieve linear complexity in the horizon. The resulting solver produces collision-free trajectories for a 7-DOF arm in milliseconds, running 100–530x faster than recent AGHF solvers with equal or better success rates, and solves reaching tasks on a 22-DOF humanoid in roughly 14 milliseconds where state-of-the-art optimizers require seconds to minutes. Extending the framework through contact, the same solver synthesizes periodic walking and stair-climbing gaits for the Talos humanoid, cold-started from a static pose and respecting friction-cone and actuation limits, in under 300 milliseconds per gait. Finally, I describe a GPU batch formulation that evaluates thousands of heat-flow solves in parallel, at nearly 3,000 solves per second on perception-derived point-cloud scenes, and its use as a differentiable layer that connects trajectory optimization to learned planning. Together, these results turn the geometric heat flow into a practical, structure-preserving engine for planning and control of complex hybrid robotic systems.

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